Swetna Annette Tribhuvan

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Education

Rochester Institute of Technology *August 2021-December 2024*
Master's in Computer Science: Minor in Big Data Analytics & Artificial Intelligence Rochester,NY

University of Mumbai *June 2017- June 2021*
Bachelor's in Information Technology Mumbai,India

Professional Experience

Information & Technology Services, RIT *May 2023 - Decemeber 2024*
Software Engineer , Intern Rochester,NY

- Engineered a **Grafana dashboard** integrated with a **Kotlin backend**, boosting **operational efficiency** by 93% and delivering real-time visualization for 78% of RIT classroom data
- Achieved 96% accuracy in energy calculations with **PostgreSQL** and **Prometheus**, enhancing **monitoring precision**
- Transformed websites using **Laravel**, **Vue.js**, and **Drupal (9.0 and 10.0)**, elevating **user satisfaction** and ensuring **system reliability**
- Refined **JavaScript** and **PHP** code, enhancing functionality and performance by 88%, while also optimizing backend efficiency
- Championed team progress and fostered innovation by leading **Scrum activities** and facilitating **collaboration**

Covid India National Taskforce *May 2021 - July 2021*
Web developer, Intern Mumbai,India

- Developed a **Django-based web interface**, streamlining data access by 22% during the COVID-19 crisis
- Enhanced **MySQL queries**, amplified performance metrics by 11%, integrated **APIs** to expand functionality by 32%
- Propelled team efficiency by 30% through implementing **Agile methodologies**, including sprints, stand-ups, and retrospectives
- Demonstrated quick learning abilities, mastering new tools and technologies within 20% less time than expected

Projects

AI-Based Traffic Flow Optimization Using Cloud Computing and Google Maps API *May 2024 – December 2024*
https://github.com/Swetna/TrafficFlowOptimization

- Devising AI-driven models leveraging **Google Maps API** and **cloud computing**, cutting congestion by up to 25% in SUMO simulations
- Expanded simulation capabilities by 40% through analyzing **large-scale traffic datasets** (over 1 million data points) on **AWS** cloud platforms
- Advancing **real-time adaptive** traffic control by 30% with cloud-based AI algorithms

Luxurious Stay: MERN Project with Enhanced UX & Security *July 2023 – August 2023*
https://github.com/Swetna/HotelBookingApp

- Designed a **MERN**-based hotel booking platform, boosting **user engagement** by 25% with a revamped interface
- Strengthened **platform security**, cut vulnerabilities by 30%, and accelerated response times by 40% through optimized integration
- Fortified scalability with **optimized database queries** and **load balancing**, supporting 50% more concurrent users

Discerning and Categorizing Poison Ivy: An Elegant Exploration *September 2022 – December 2022*
https://github.com/Swetna/PoisonIvyDetection

- Pioneered an **analysis system** using **advanced methods** to identify poison ivy, enhancing outdoor safety initiatives
- Curated a dataset of **1,000+ images**, achieving 89% accuracy in plant identification
- Elevated model performance by 15% through **data augmentation** and **transfer learning**

For more projects, visit portfolio: https://swetna.github.io/SwetnaTribhuvanPortfolio/

Technical Skills

- **Languages**: Java, Python, JavaScript, SQL, PHP, C#, XML, C++, Matlab, C, R, HTML, CSS
- **Frameworks**: Grafana, Kubernetes, Django, Spring Boot, React, NodeJS, .NET, Drupal, Flask, REST API, phpMyAdmin, Kotlin, Laravel, Vue.js
- **Cloud/Database**: DBMS, PostgreSQL, MongoDB, MySQL, Redis, SimpleDB, DynamoDB, Prometheus, SAAS
- **ML & DevOps Tools**: Scikit Learn, TensorFlow, NumPy, Pandas, Matplotlib, Amazon Web Services (AWS) Google Cloud, Docker, Git, JIRA , Jenkins
- **Concepts/Interests**: Object-Oriented Programming (OOP), Algorithms & Data Structures, Android Development, Agile, Scrum
- **Soft Skills**: Communication, Leadership, Time Organization, Teamwork

Certifications

- **AWS Certified Developer Associate** Amazon Web Services *December 2024*
- **C++ Essential Training** LinkedIn learning *July 2024*
- **AWS Certified Cloud Practitioner** Amazon Web Services *April 2024*
- **Introduction to Generative AI** Google Cloud *July 2023*